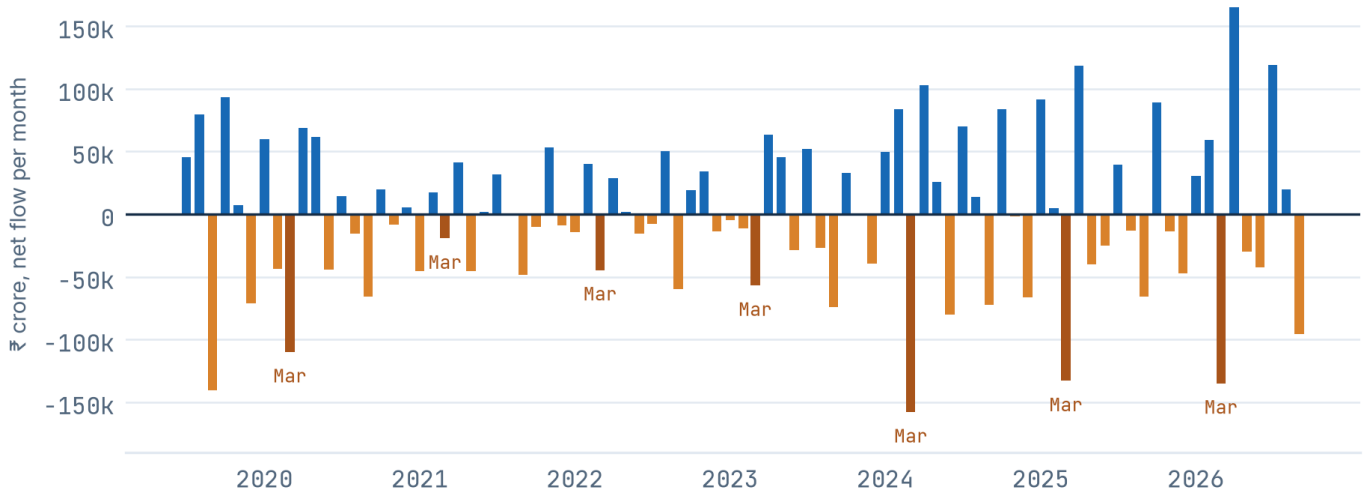
भारतीय प्रौद्योगिकी संस्थान पटना
Indian Institute of Technology Patna

Zeckathon 2026
Problem Statement 03

LIQUIDFLOW

# How much cash should a liquid fund hold?

Forecasts to March 2027 of money moving in and out of India's liquid and overnight funds, and a cash-buffer rule built on them.




Net flow into liquid funds each month, July 2019 to September 2026, in ₹ crore. Blue is money in, orange is money out. Every March, the darkest bars, has been an outflow.

> **In short:** liquid funds should hold 3 to 9% of assets in an ordinary month, 21% going into December and 31% going into March, when we expect about ₹1.16 lakh crore to leave.

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Submitted 29 October 2026 · Data to September 2026

SUMMARY

# The short answer

Liquid funds should keep 3 to 9% of assets ready to pay out in an ordinary month, about 21% going into December and about 31% going into March 2027. March is the hardest month of the year: in each of the last three, 23% to 31% of the category's money left.

**31.3%**
Liquid-fund buffer for March 2027, about ₹1.96 lakh crore

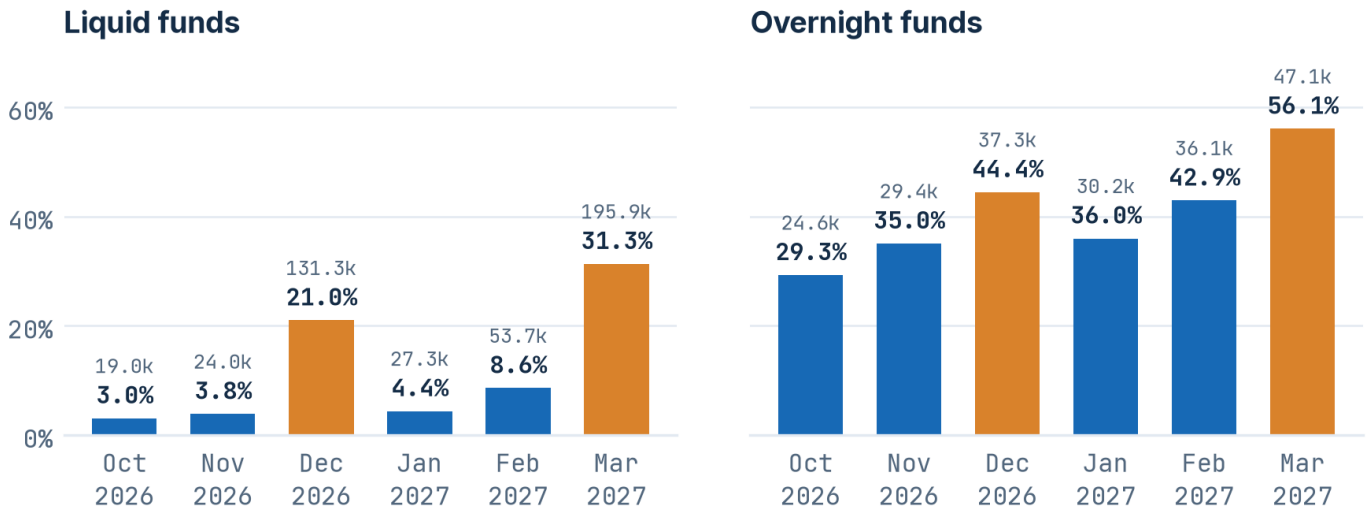
**7 of 7**
Marches since 2020 with net outflows from liquid funds

**4 of 48**
Backtest months where our buffer fell short. Last year's rule: 18

**87.5%**
Lower shortfall cost than last year's rule, on the brief's cost formula

## Cash to hold, month by month

Buffer as a share of September 2026 assets, with the amount in thousand ₹ crore above each bar. It is set so that only a 1-in-20 month should need more. Orange marks advance-tax months.




## What a fund manager should take from this

01 **Plan around the tax calendar.** Advance tax is due on 15 June, 15 September, 15 December and 15 March. Companies redeem to pay it, so those months carry the big outflows. The calendar was the most useful input in our models.

02 **Size cash from the downside, not the average.** November's central forecast is an inflow of ₹47,097 crore, yet there is a 1-in-20 chance of an outflow above ₹23,989 crore. The buffer covers that tail.

03 **Overnight funds need proportionally more.** They are a seventh the size of liquid funds and swing harder, so their buffer runs at 29% to 56% of assets.

WHAT DRIVES THE FLOWS

# Money leaves on tax dates

Most of the money in liquid funds belongs to companies parking short-term cash. When advance tax falls due they take it back, and the same rhythm shows up year after year. It is sharpest in March, when the fourth instalment lands on top of the financial year-end.

## Liquid-fund net flow as % of the previous month's assets

Boxed columns are advance-tax months. The bottom row counts the years in which the month had a net outflow.

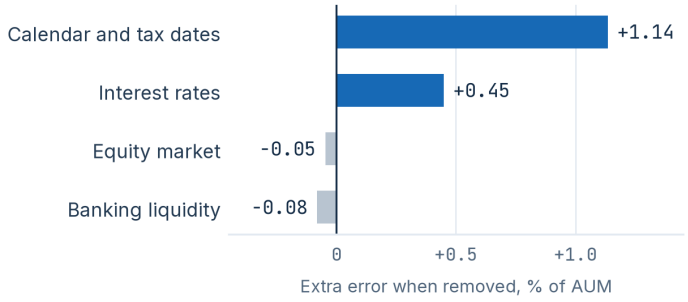
| | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2019 | | | | | | | | +18 | -27 | +24 | +1 | -14 |
| 2020 | +14 | -9 | -25 | +21 | +15 | -9 | +3 | -4 | -15 | +5 | -2 | +1 |
| 2021 | -12 | +5 | -5 | +12 | -12 | +1 | +9 | 0 | -13 | -3 | +17 | -2 |
| 2022 | -4 | +12 | -11 | +8 | 0 | -4 | -2 | +14 | -15 | +5 | +9 | -3 |
| 2023 | -1 | -3 | -15 | +19 | +11 | -6 | +12 | -6 | -16 | +9 | 0 | -10 |
| 2024 | +13 | +19 | -31 | +28 | +6 | -16 | +17 | +3 | -14 | +19 | 0 | -13 |
| 2025 | +20 | +1 | -23 | +27 | -7 | -5 | +8 | -2 | -12 | +19 | -3 | -9 |
| 2026 | +6 | +11 | -23 | +35 | -5 | -7 | +21 | +3 | -13 | | | |
| Outflow years | 3/7 | 2/7 | **7/7** | 0/7 | 3/7 | **6/7** | 1/7 | 4/8 | **8/8** | 1/7 | 4/7 | **6/7** |

## March, year by year

| Year | Net flow, ₹ cr | % of Feb assets |
|---|---|---|
| 2020 | -110,037 | **-24.8%** |
| 2021 | -19,384 | **-5.5%** |
| 2022 | -44,604 | **-11.5%** |
| 2023 | -56,924 | **-14.7%** |
| 2024 | -157,970 | **-30.5%** |
| 2025 | -133,034 | **-23.4%** |
| 2026 | -134,988 | **-22.5%** |

A repeat of the last three Marches at September 2026 size would mean ₹1.4 to 1.9 lakh crore leaving the category in March 2027. Our forecast is a ₹1.16 lakh crore outflow, with a buffer of ₹1.96 lakh crore (page 4).

## Which inputs helped the forecast

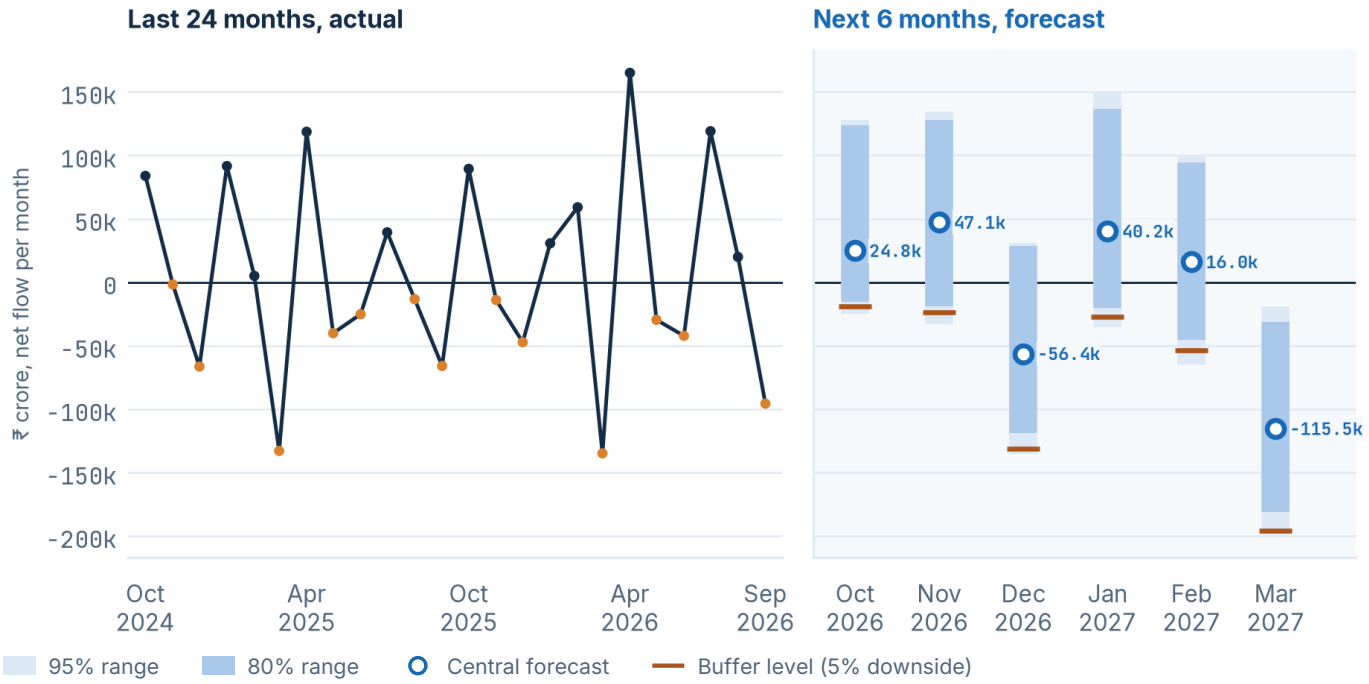



Extra forecast error when each input group is removed from our Ridge model, measured on the Oct 2023 to Sep 2024 selection year. Calendar and interest rates carried the signal. Equity and banking-liquidity data did not add to it.

FORECAST · CRORE RUPEES

# Liquid funds: the next six months

We expect money to come in during October, November, January and February, and to go out in the two advance-tax months: about ₹56,397 crore in December and ₹115,546 crore in March. March is the only month where even the top of the 95% range is an outflow. The buffer is set from the bottom of the range, not from the central forecast.




| Month | Central | 80% range | 95% range | Buffer | % of assets |
|---|---|---|---|---|---|
| Oct 2026 | 24,824 | -15,519 to 123,635 | -24,671 to 128,096 | **19,025** | 3.0% |
| Nov 2026 | 47,097 | -19,245 to 127,959 | -33,387 to 134,578 | **23,989** | 3.8% |
| Dec 2026 | -56,397 | -118,676 to 28,780 | -135,630 to 31,071 | **131,287** | 21.0% |
| Jan 2027 | 40,212 | -20,274 to 136,483 | -35,650 to 149,521 | **27,262** | 4.4% |
| Feb 2027 | 15,976 | -45,448 to 94,421 | -65,054 to 98,862 | **53,653** | 8.6% |
| Mar 2027 | -115,546 | -181,514 to -31,397 | -199,025 to -19,076 | **195,924** | 31.3% |

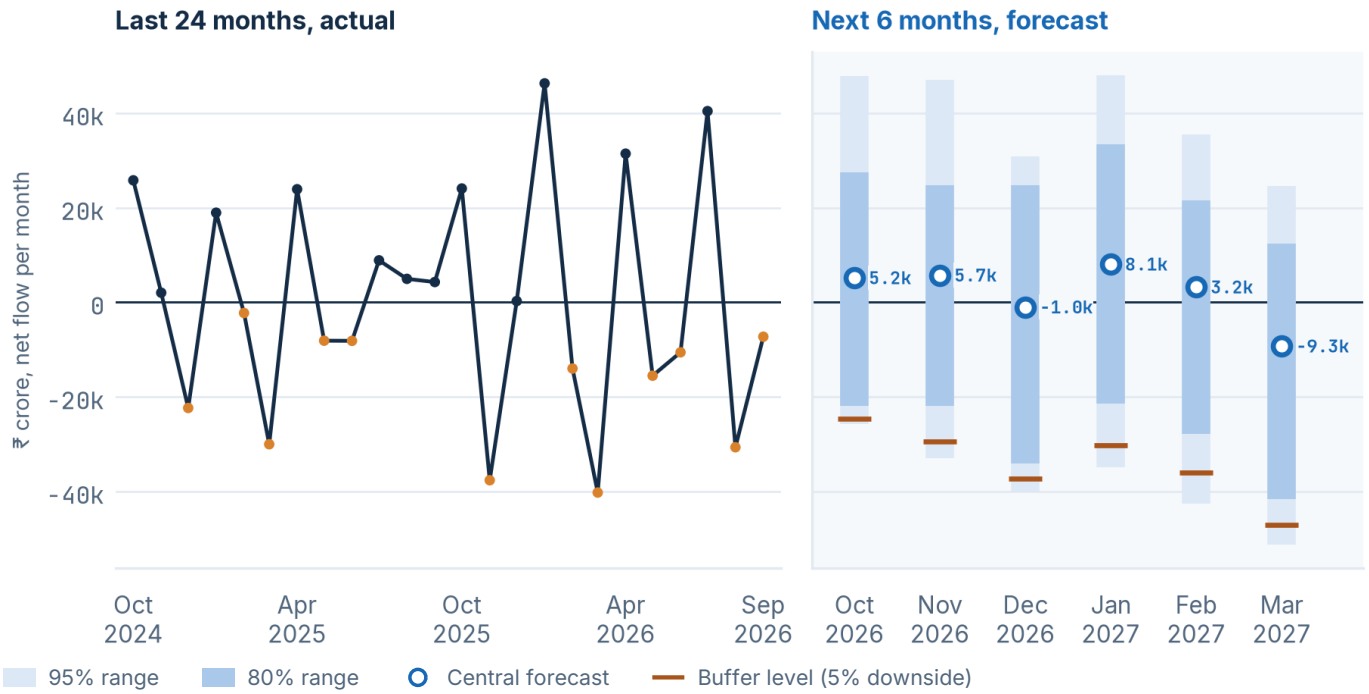
October is a nowcast: AMFI publishes October figures in early November. January to March are four to six months ahead, beyond the brief's three; they use the same model and range method and were backtested on the same two years. Buffers are per month and should not be added together.

**How sure are we?** Over the two-year test, the liquid-fund 95% range caught 93% of actual outcomes one to three months ahead and 97% four to six months ahead. The 80% range caught 71% and 72%, so the inner band runs a little narrow. The one-month buffer fell short in 1 of 24 months (page 6).

FORECAST · CRORE RUPEES

# Overnight funds: the next six months

Overnight funds look close to flat: small inflows in October, November, January and February and small outflows in December and March. The category is small, about ₹84,033 crore, so a ₹30,000 crore swing is a third of it. That is what pushes the buffer to 29% to 56% of assets.




| Month | Central | 80% range | 95% range | Buffer | % of assets |
|---|---|---|---|---|---|
| Oct 2026 | 5,176 | -21,916 to 27,489 | -25,764 to 47,871 | **24,593** | 29.3% |
| Nov 2026 | 5,734 | -21,869 to 24,761 | -32,964 to 46,996 | **29,448** | 35.0% |
| Dec 2026 | -1,007 | -34,140 to 24,687 | -39,998 to 30,810 | **37,346** | 44.4% |
| Jan 2027 | 8,057 | -21,400 to 33,460 | -34,954 to 48,029 | **30,230** | 36.0% |
| Feb 2027 | 3,220 | -27,831 to 21,599 | -42,579 to 35,536 | **36,069** | 42.9% |
| Mar 2027 | -9,259 | -41,595 to 12,385 | -51,265 to 24,528 | **47,102** | 56.1% |

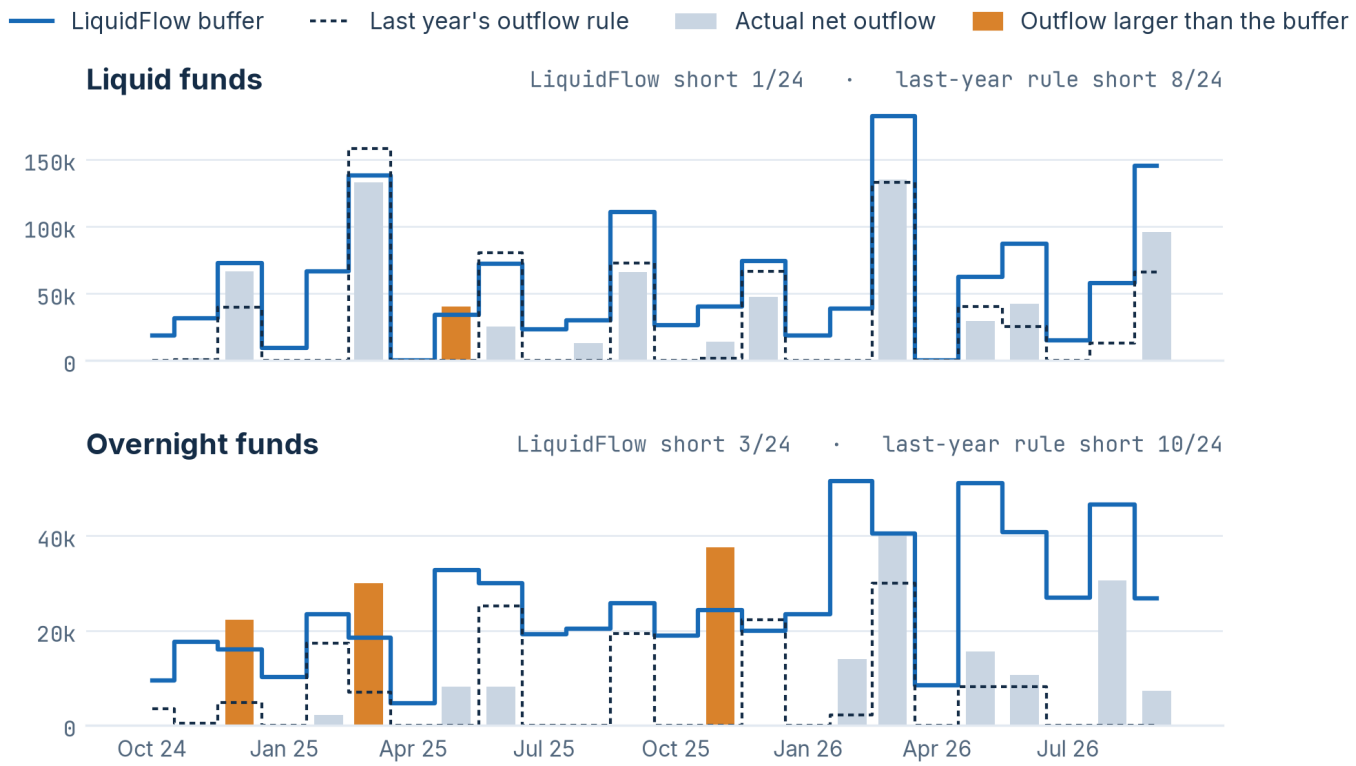
October is a nowcast: AMFI publishes October figures in early November. January to March are four to six months ahead, beyond the brief's three; they use the same model and range method and were backtested on the same two years. Buffers are per month and should not be added together.

**How sure are we?** Less sure than for liquid funds. Over the two-year test, the overnight 95% range caught 78% of actual outcomes one to three months ahead and 78% four to six months ahead, and the 80% range 68% and 58%, so both bands are too narrow. The one-month buffer fell short in 3 of 24 months (page 6); treat it as a minimum.

BACKTEST · OCT 2024 TO SEP 2026

# The buffer, tested on two years

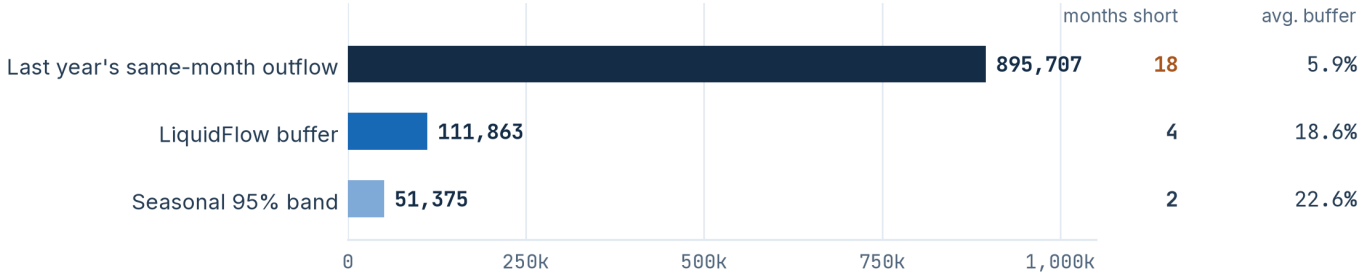
We replayed October 2024 to September 2026 one month at a time. Each buffer was set using only what was known that month, then compared with what actually left. The benchmark is the rule named in the brief: hold last year's outflow for the same month.




₹ crore, one month ahead, 24 months per category. Bars show net outflow; months with net inflow have no bar.

## What each rule costs

Cost from the brief: 0.5% a year on cash held beyond what was needed, plus ₹3 for every rupee of shortfall, summed over all 48 category-months, in ₹ crore.




LiquidFlow cut the cost by 87.5% against last year's rule. It holds more cash, 18.6% of assets on average against 5.9%, and in return it ran short in 4 months instead of 18. A plain seasonal 95% band, which holds even more (22.6%), was cheaper again. Most of the saving comes from sizing cash to the downside of a forecast range, and that idea holds across both methods.

METHOD

# How LiquidFlow works

**01 Data**
87 monthly AMFI reports per category, plus RBI rates, Nifty 50 and liquidity data

→

**02 Inputs**
Calendar and tax dates, interest rates, equity and banking liquidity

→

**03 Models**
Seven candidates, one chosen on the 2023 to 2024 year, then locked

→

**04 Ranges**
Built from the model's own last 24 errors

→

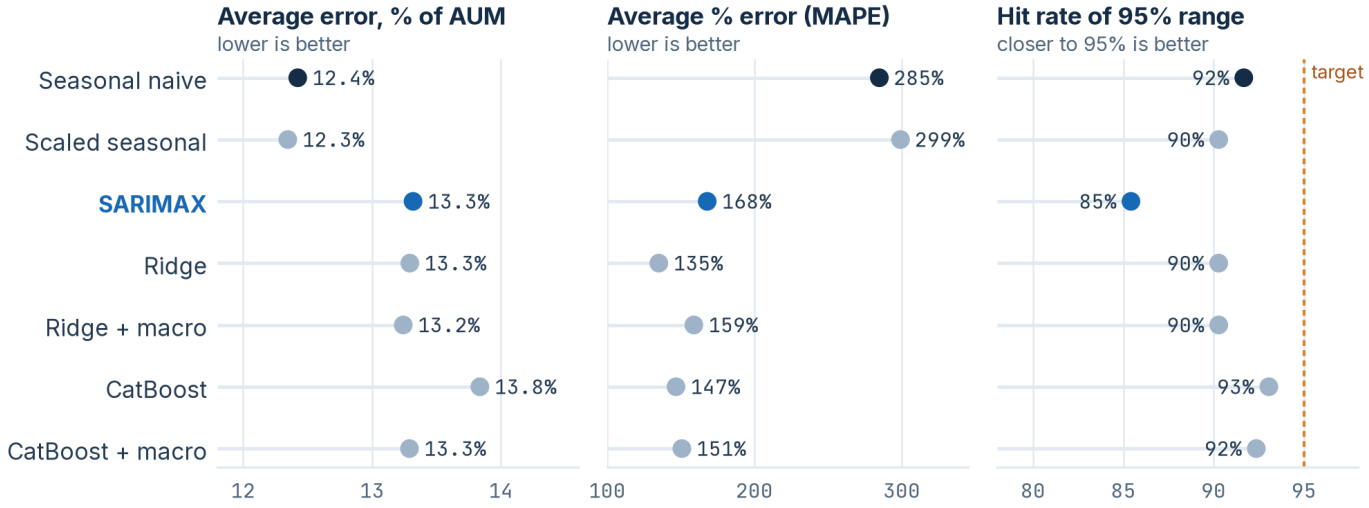
**05 Buffer**
Cash to cover the 5% downside, capped at assets

## A fair test

We chose the model on one year of data, October 2023 to September 2024, and locked that choice before looking at the two test years. Each forecast uses only data that was public at the time: economic inputs lag by two months, and an error only feeds the ranges once its month has been reported. All seven candidates got the same treatment.

## How the seven models compared on the test years

SARIMAX, in blue, was chosen in the selection year. The seasonal naive benchmark, in navy, repeats last year's figure for the same month.




Simple seasonal methods were slightly more accurate in rupee terms (12.4% of assets against 13.3% for SARIMAX), while SARIMAX cut percentage error by 41%. The machine-learning models, Ridge and CatBoost, landed close to SARIMAX and did not beat the seasonal methods on rupee error. With monthly data, the calendar does most of the work, and a seasonal method already captures it. Better data is a surer way forward than a bigger model.

DEFINITIONS · SCOPE · REFERENCES

# Notes and sources

## Definitions

**Net flow** is subscriptions minus redemptions for the month, taken straight from AMFI's reported columns.

**Buffer** = `max(0, -q05)`, where `q05` is the 5th percentile of forecast net flow, capped at the category's assets.

**Cost** = idle-cash cost + shortfall penalty, where `O` is the actual net outflow and `B` the buffer:

```
0.5% × days/365 × max(B - O, 0)
+ 3 × max(O - B, 0)
```

If the penalty is read as three times the monthly yield gap instead of ₹3 per rupee, holding little cash becomes the cheaper choice and last year's rule edges ahead (₹461 crore against ₹487 crore). We follow the brief's wording.

## Using this for one fund

These forecasts cover whole categories, month by month. A single fund planning for 15 March also needs its own daily redemptions, how concentrated its investors are, and when its holdings mature.

Timing matters within the day. Redemptions are paid on the morning of T+1, while money from TREPS and reverse repo comes back in the evening.

Liquid funds may hold instruments of up to 91 days. The 20% eligible-liquid-assets rule is not a 20% cash requirement.

With fund-level daily data, the same method applies: forecast the range, then hold cash against its downside.

## Data

| Source | Coverage | Used for |
|---|---|---|
| AMFI monthly reports | Jul 2019 to Sep 2026 | Net flows and assets, liquid and overnight |
| RBI call-money rate | Apr 2020 to Jun 2026 | Interest-rate inputs |
| Nifty 50 monthly average (RBI) | Apr 2018 to Jun 2026 | Equity-market input |
| RBI liquidity operations | Jul 2019 to Jul 2026 | Banking-liquidity input |
| Income Tax Department | Advance-tax schedule | Calendar inputs |

## References

[1] Zeckathon PS03 problem brief
[2] AMFI monthly data archive
[3] RBI Handbook of Statistics, Tables 174 and 182
[4] RBI Bulletin, liquidity operations
[5] Income Tax Department, advance-tax schedule
[6] SEBI Master Circular for Mutual Funds, March 2026
[7] Hyndman and Athanasopoulos, Forecasting: Principles and Practice

**Team Angry Bird**, IIT Patna: Jayant Raj, Rohan Kumar Singh, Harshit Kumar and Viraj Singh. The project repository holds the code, data snapshots and an interactive dashboard. Running `make all` checks the source files, reruns the experiment and rebuilds every figure and number in this report.